

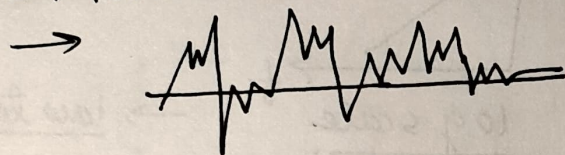
Introduction to Jitter & Phase Noise

* Jitter & Phase noise

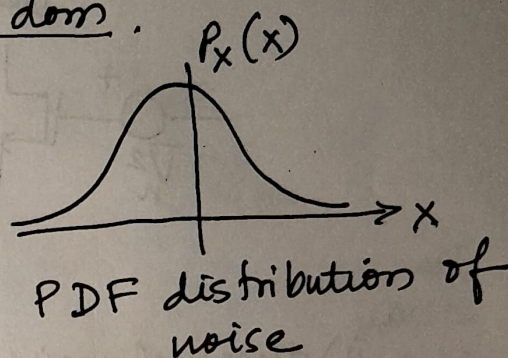
* Osc need to be designed only after having phase noise

Noise in frequency domain & time dom.

→ Noise is random.



noise



→ time domain
→ Gaussian distri

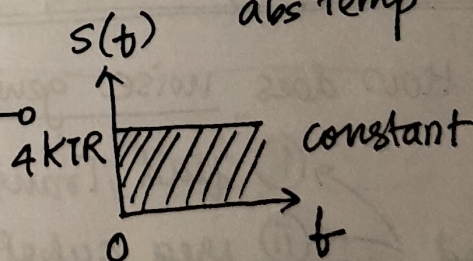
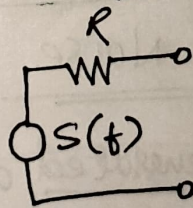
Noise in frequency domain.

→ PDF: $S(f)$ → how power is present in a 1 Hz BW around each frequency.

2.1.2 Device Noise.

① Resistor Noise

converts thermal to voltage
 $S(f) = 4KTR$ V^2/Hz
Boltzman const
abs temp



Example 2.1

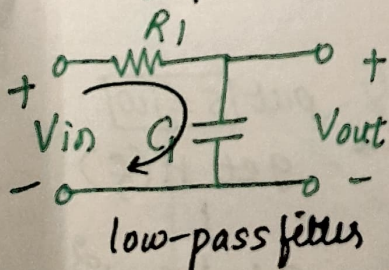
o/p noise of RC filter?

$$s(f) = s_{n1}(f) \cdot |H(f)|^2$$

$$H(f) = (R_1 C s + 1)^{-1}$$

$$\therefore s_{n,out}(f) = 4KTR \cdot \frac{1}{R_1^2 C^2 (\omega^2 + 1)^2}$$

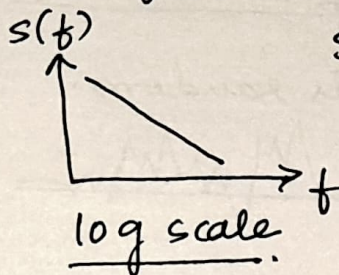
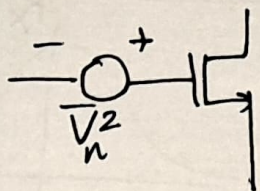
in V^2/Hz



- ② MOSFET noise.
- flicker noise
 - thermal noise

i) flicker noise. ($1/f_{\text{noise}}$)

→ modeled as voltage source in series with the gate



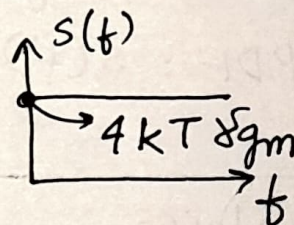
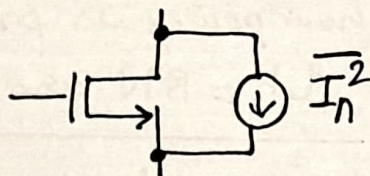
$$S_{1/f}(f) = \frac{K}{WL C_{ox} f} V^2 / \text{Hz}$$

→ low temp.

→ reduced by increasing $(W \times L)$

ii) thermal noise.

- in Sat region
- modeled as current source b/w drain & source



→ flat noise

2.1.3 Propagation of Noise

→ How does noise generated at source affects output?

- Method steps
- ① find Transfer f^n
 - ② freq substitution
 - ③ output Spectrum

④ TF

→ get $H(s)$ → from V_{n1} to $V_{n,out}$ →
(as if noise is normal)

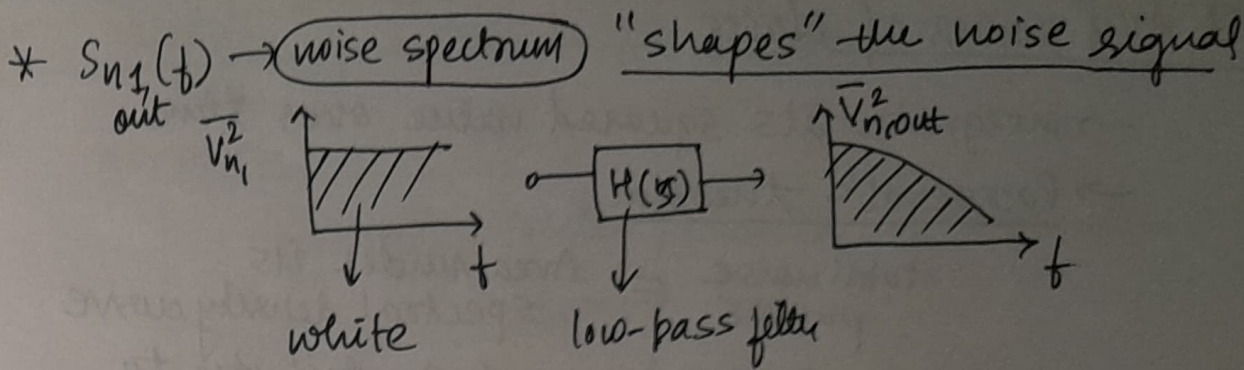
put $[s=j\omega]$

get $H(f)$

↓

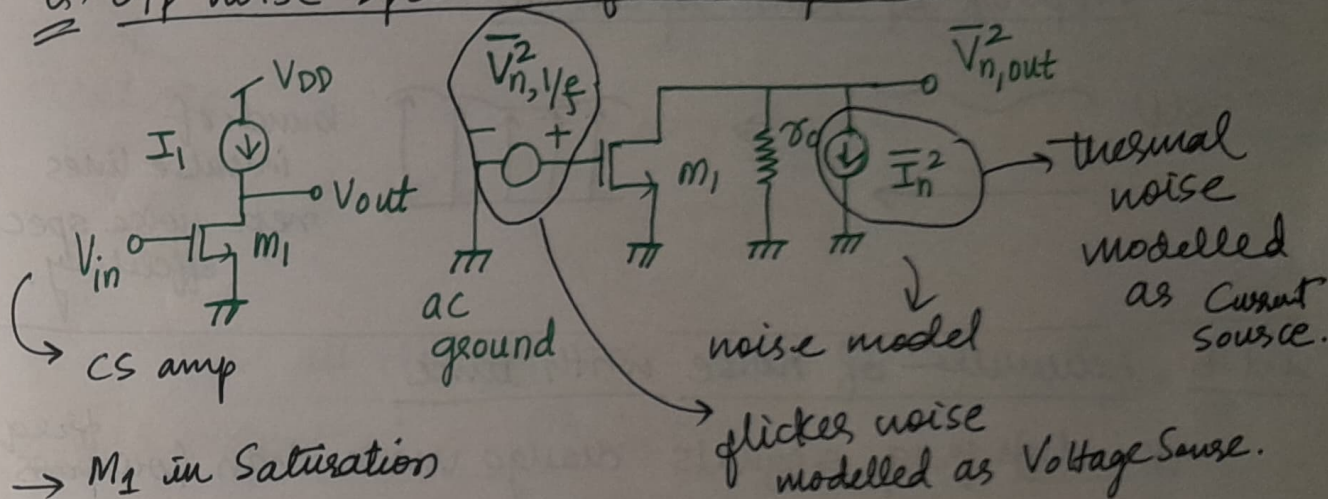
get $|H(f)|^2$

o/p noise ← $S(f) = S_{n1}(f) \cdot |H(f)|^2$



Example 2.3.

Q. o/p noise spectrum of CS amp stage



$\rightarrow$ M_1 in Saturation

Transfer functions

1/f noise: passes thro voltage gain of the stage.

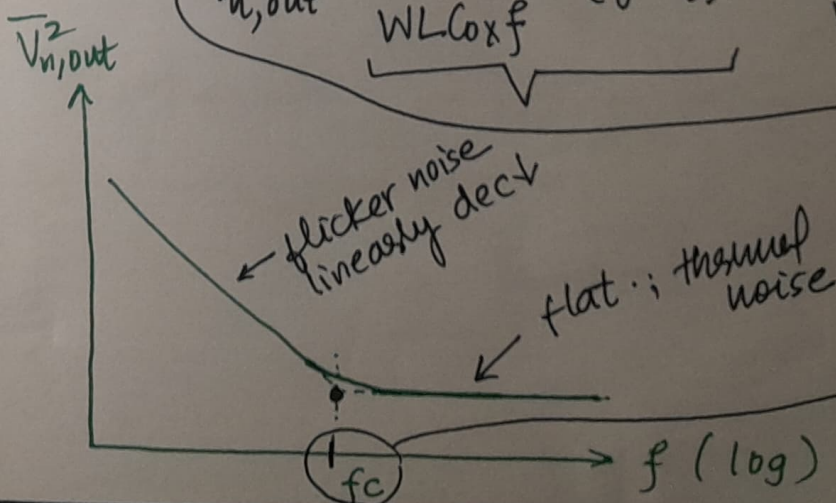
thermal noise:

$$g_m r_o \rightarrow o/p = (g_m r_o)^2$$

drain I passes thro the o/p as resistance $r_o \rightarrow$ contribution is $(r_o)^2$

O/p Noise Spectrum:

$$\overline{V_{n,out}^2} = \frac{K}{WL C_{ox} f} \cdot (g_m r_o)^2 + \underbrace{4kT r f g_m (r_o)^2}_{\text{thermal noise}}$$



Corner freq (f_c)

flicker noise is dom at low freq.

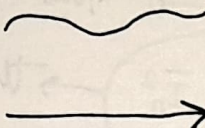
2.1.4 Avg power of Noise

→ integrate its squared value over time

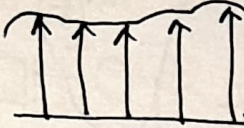
→ Parseval's theorem

total noise power = Area under its spectral density curve
✓ every freq component contributes to time-domain fluctuations.

2.1.5 Approx of Noise Spectrum

$s(t)$ 

$\Rightarrow$



bundle of
impulse lines
rep. noise spec.
efficiency.

2.1.6 Accumulation of noise with time

* High freq signals change more than low ~~pass~~ ^{freq} in short observation windows.

$\approx$

BASIC Jitter & Phase noise Concepts

2

noiseless oscillators

- ✓ perfect strictly periodic oscillations
- T_1 : period doesn't change
- frequency is constant
- Zero crossings occur at predictable intervals

$$t = n \frac{T_1}{2}$$

Real osc.

noise disturbs

- amplitude
- frequency
- phase

✓ all changes

T, f , zero crossings change.

Excess phase (noise induced)

noisy output of oscillator →

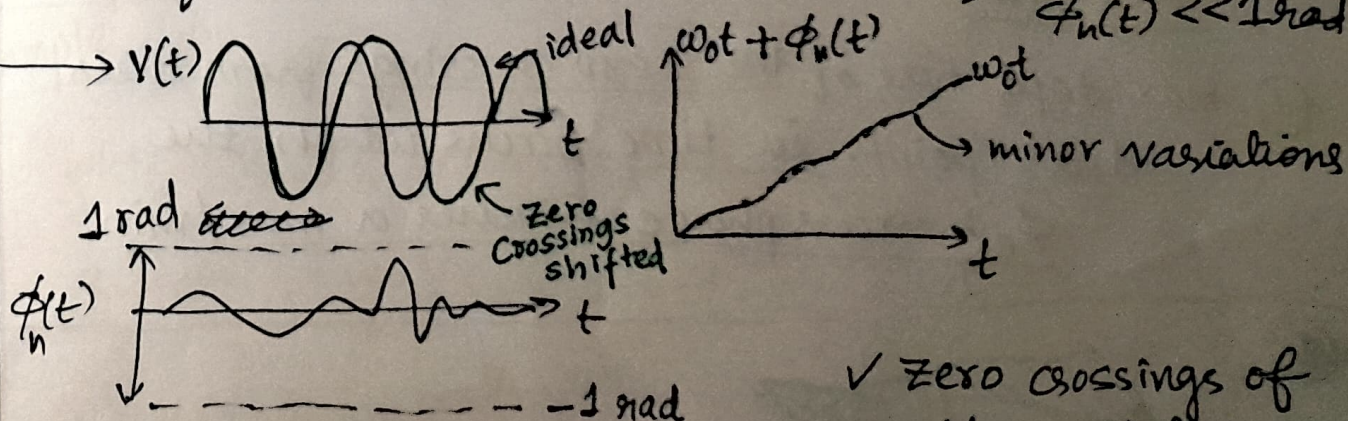
~~Vout(t)~~

$$V_{out}(t) = V_0 \cos(\omega_0 t + \phi_n(t))$$

ω_0 : carrier frequency (ideal) no noise

Example 2.4. Plot total phase & excess phase

for $V_{out}(t) = V_0 \sin(\omega_0 t + \phi_n(t))$ if $\phi_n(t) \ll 1 \text{ rad}$



✓ Zero crossings of V_{out} shifted due to noise

2-2.1 Jitter → def deviation of zero crossings (or ed) of a periodic signal from ideal points.

✓ time-domain phenomenon.

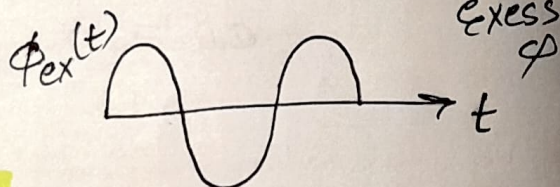
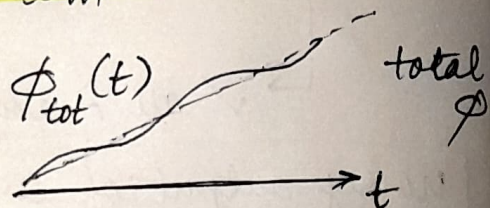
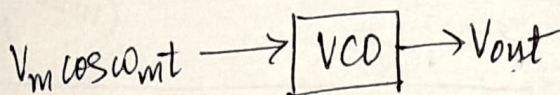
Example 2.5: VCO exhibit jitter ???

given: $V_{\text{cont}}(t) = V_m \cos \omega_m t$ ← input

$$V_{\text{out}}(t) = V_o \cos(\omega_o t + K_{\text{vco}} \int V_m \cos \omega_m t \cdot dt)$$

$$V_{\text{out}}(t) = V_o \cos\left(\omega_o t + \frac{K_{\text{vco}} V_m}{\omega_m} \sin \omega_m t\right)$$

✓ output



peak to peak
Jitter $\Rightarrow \frac{2 K_{\text{vco}} V_m}{\omega_m}$ rad

Imp

: Observations:

- ① max departure of the zero crossings from their ideal points in time occurs when the Φ_{ex} excess phase reaches a max/min.

Deterministic Jitter (DJ)

if the oscillation frequency is periodically
due to explicit Voltage Control disturbed.

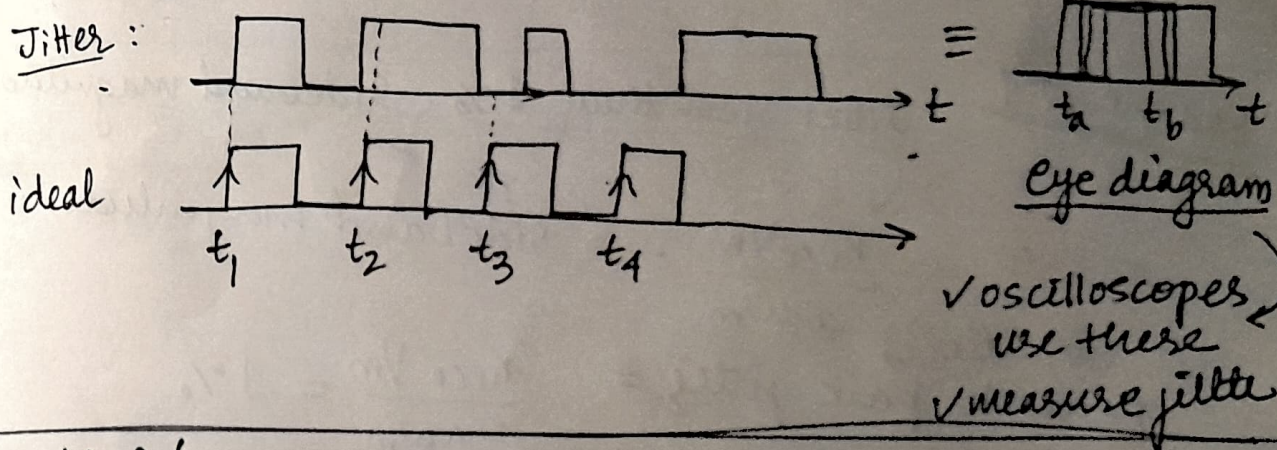
$$\checkmark V_m \cos(\omega_m t) \rightarrow \boxed{\text{VCO}} \rightarrow V_{out}$$

$$\checkmark V_{out} = V_0 \cos \left\{ \omega_0 t + \frac{K_{vco} V_m \sin(\omega_m t)}{\omega_m} \right\}$$

• peak to peak Jitter = $\frac{2 K_{vco} V_m}{\omega_m} \text{ rad} = \frac{2 K_{vco} V_m}{\omega_m \omega_0} \text{ in seconds}$

✓ Zero crossings shift periodically ✓

Eye diagram



sample 2.6

$$V_m \cos(\omega_m t) \rightarrow \boxed{\text{VCO}} \rightarrow \text{o/p ???}$$

wkt. $V_{out}(t) = V_0 \cos \left(\omega_0 t + \frac{K_{vco} V_m \sin(\omega_m t)}{\omega_m} \right)$

assume $\frac{K_{vco} V_m}{\omega_m} \ll 1 \text{ rad.} \Rightarrow \sin \theta \approx \theta$

& $\cos \theta \approx 1$ for $\theta \ll 1$

$V_{out}(t) \approx V_m \cos(\omega_0 t) - V_0 \frac{K_{vco} V_m}{\omega_m} \sin \omega_0 t \sin \omega_m t$
 $\approx V_m \cos(\omega_0 t) - V_0 \left(\frac{K_{vco} V_m}{\omega_m} \right) [\cos(\omega_0 - \omega_m)t - \cos(\omega_0 + \omega_m)t]$
"narrowband FM approximation"

$$\Rightarrow V_{out} \approx V_0 \cos \omega_0 t - V_0 \frac{K_{VCO} V_m}{2\omega_m} [\cos(\omega_0 - \omega_m)t - \cos(\omega_0 + \omega_m)t]$$

✓ main lobe at ω_0 ✓

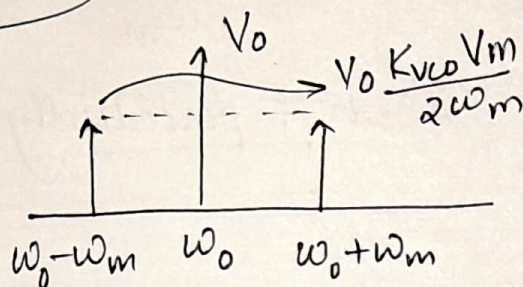
✓ side lobes at $(\omega_0 + \omega_m)$ & $(\omega_0 - \omega_m)$

✓ sidebands' magnitude w.r.t the main carrier mag

$$= \boxed{\frac{K_{VCO} V_m}{2\omega_m}}$$

as $\omega_m \uparrow \uparrow$
Jitter reduces

$\omega_0 \pm \omega_m$ → show jitter



Example 2.7.

Jitter less than 1%. Sideband magnitude

$\frac{K_{VCO} V_m}{2\omega_m}$ → sideband magnitude

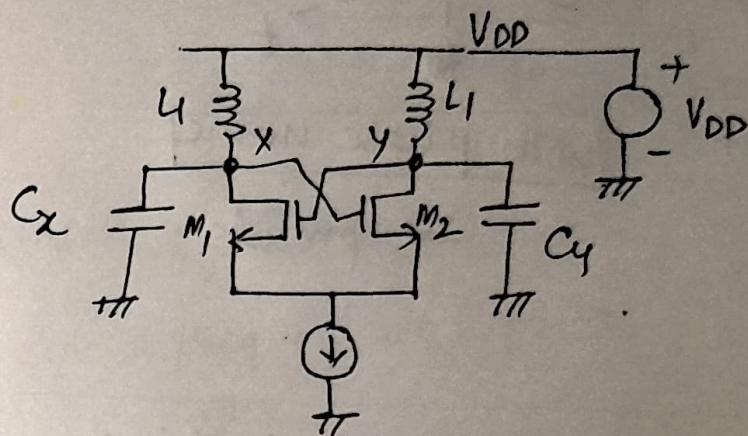
normalised peak-to-peak jitter = $\frac{K_{VCO} V_m}{\pi \omega_m} = 1\%$

$$\frac{K_{VCO} V_m}{2\omega_m} = \frac{\pi}{200}$$

$$= \underline{\underline{-36\text{dB}}}$$

Example 2.8 LC oscillator o/p sideband mag?

supply noise $\Rightarrow V_{DD} = V_{DD0} + V_m \cos \omega_m t$



$\rightarrow$ fluctuations in V_{DD} cause o/p freq

$\rightarrow$ o/p frequency $\omega_1 = \frac{1}{\sqrt{L_1 (G + CDB)}}$

$\rightarrow$ CDB varies with V_{DD}

$\rightarrow$ Gain: K_{VCO} from V_{DD} to o/p freq

CDB =

(*) normalised mag of each sideband = $\frac{K_{VCO} V_m}{2 \omega_m}$ ✓

2.2.2 Phase Noise

def random fluctuations in the phase (or zero crossings) of an osc o/p.

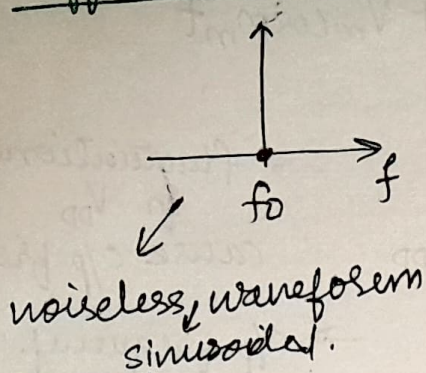
✓ jitter can add both random & deterministic variations (periodic)

but phase noise is concerned with random variations only.
 $\downarrow$
 in frequency domain

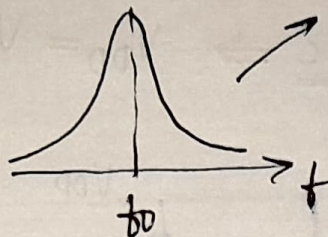
$$V_{out}(t) = V_o \cos(\omega_o t + \phi(t))$$

$\rightarrow$ phase noise.

Effect on Spectrum.



ideal → impulse



with phase noise

→ spreads

Narrowband approx for phase fluctuations.

$$V_{out}(t) = V_0 \cos(\omega_0 t + \phi_n(t))$$

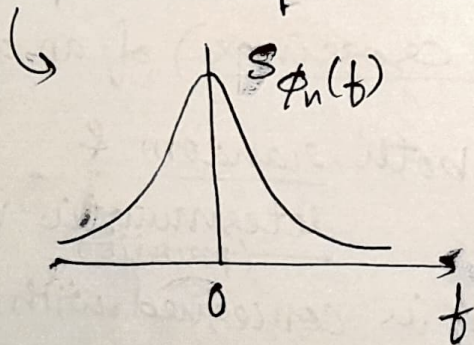
$$V_{out}(t) = \underbrace{V_0 \cos \omega_0 t}_{\text{main carrier}} - \underbrace{V_0 \phi_n(t) \sin \omega_0 t}_{\text{phase noise side}}$$

main carrier

phase noise side

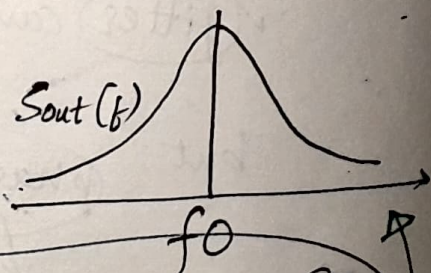
Spectral Relationship

phase noise spectrum $\phi_n(t)$



output voltage spectrum

$S_{out}(f)$



NOTE : ①

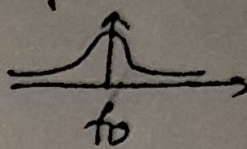
$\phi_n(t)$ spectrum is shifted to f_0 in the output voltage spectrum

②

farther you go from f_0 (i.e., larger offset) less likely you are to see large deviations - spectrum tapers off.

✓ tapering gives BELL Shape around f_0

Basically due to noise in phase, $\uparrow$ cleanspike
at f_0 into a shape that is ^{to} centered
around f_0

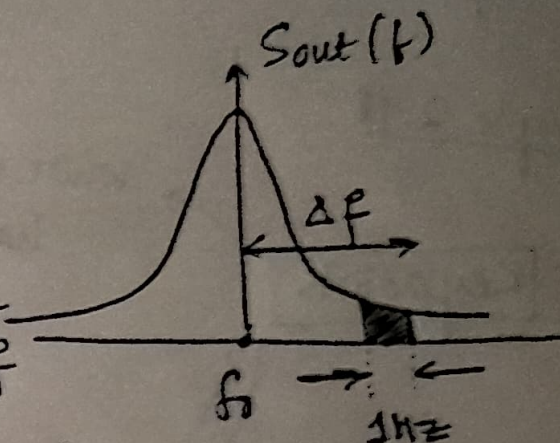


Measurement of Phase Noise.

(a) Select a freq offset (Δf) .

Ex: 100kHz

(b) Calculate the noise power
in a 1Hz BW at Δf



(c) Normalise this noise power to
carrier power ($V_0^2/2$)

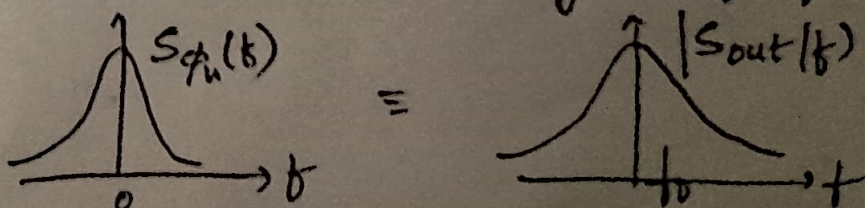
(d) Express in dBc/Hz: decibels below the carrier,
per hertz.

Imp

A good VCO for GSM at 900MHz may
require phase noise better than -145dBc/Hz
at a 25MHz offset.

Example 2.9.

The normalised op voltage spectrum $S_{out}(f)$ at
($f_0 + \Delta f$) is the same as the phase noise spectrum
 $S_{\phi_n}(f) \rightarrow$ after proper scaling & freq shifting



2.2.4 Relationship b/w Phase Noise & Jitter

Jitter → time domain
→ can come from deterministic or random

■ Peak-to-peak vs RMS jitter

✓ for deterministic : peak to peak jitter (in rad)
= 4 times normalised sideband mag.

✓ for random : use a rms value

phase noise → freq. domain
→ random variations

■ RMS Jitter & Phase Noise

Calculating J_{rms} from phase noise profile

$$V_0 \cos(\omega_0 t + \phi_n(t))$$

$\phi_n(t)$ → fluctuations in phase
→ time dom

→ Exhibits "Average power"

$$\therefore \text{Average power} = \overline{\phi_n^2(t)} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} \phi_n^2(t) \cdot dt$$

diff to compute

⇒ Parseval's theorem → Avg power = Area under spectrum

$$\therefore \overline{\phi_n^2(t)} = \int_{-\infty}^{+\infty} S_{\phi_n}(f) df$$

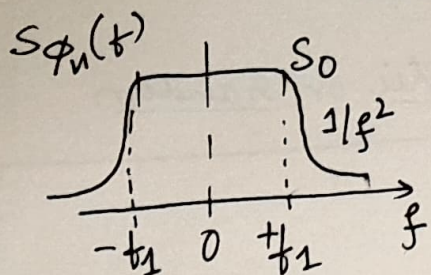
$$J_{rms} = \phi_n^2(t)$$

$$J_{rms} = \sqrt{\int_{-\infty}^{+\infty} S_{\phi_n}(f) df}$$

in radians

↗ divide by 2π & multiply by T to get in sec.

Example 2.13 phase noise of PLL is flat (S_0) for $|f| < f_1$ and falls for $|f| > f_1$ as $1/f^2$



$J_{rms} ???$

$$\rightarrow \Phi_n^2(t) = 2 \int_{-\infty}^{+\infty} S_0 df + 2 \int_{f_1}^{+\infty} S_{\phi_n}(f) df$$

$$2S_0 f_1 + 2 \int_{f_1}^{+\infty} \frac{S_0 f_1^2}{f^2} df$$

$$\Phi_n^2(t) = 2S_0 f_1 + 2S_0 f_1$$

$$\Rightarrow \boxed{\Phi_n^2(t) = 4S_0 f_1}$$

$$J_{rms} = 4 (\text{Area under Spectrum } 0 \text{ to } f_1)$$

Example 2.14 Control line of VCO: driven by R_1 = thermal noise

→ phase noise falls as $1/f^2$

→ Area is ∞ → phase grows unbound.

→ Jitter is $\infty \leftrightarrow$ RMS value = ∞ .

2.5 Types of Jitter

Jitter $\nearrow$ deterministic

Jitter $\rightarrow$ Absolute $\rightarrow$ Ever growing phase difference b/w noiseless & noisy oscillators at the same frequency.

$\downarrow$

unbound for a free-running oscillators

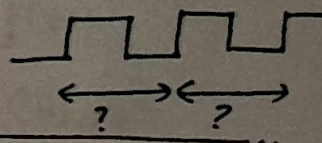
Cycle-to-Cycle Jitter (J_{cc}) $\rightarrow$ measures variation in the duration of consecutive clock cycles.

$$J_{cc} = T_k - T_{k-1}$$

$\rightarrow$ previous k^{th} clock cycle period

how much the Length of one clock period differs from the next one

- ✓ it is Bound & measurable
- ✓ measures short-time instability



$$J_{cc, rms} = \lim_{n \rightarrow \infty} \sqrt{\frac{1}{n} \sum_{k=1}^n (T_k - T_{k-1})^2}$$

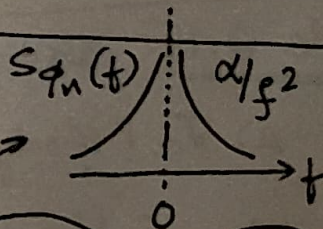
$$J_{cc} \ll J_{absolute}$$

phase noise & (J_{cc})

white noise induced phase noise $\rightarrow$

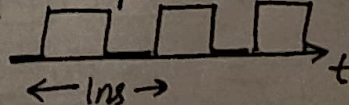
$$S_{\phi_n}(f) = \frac{\alpha}{f^2}$$

$\alpha \rightarrow$ constant



1 GHz frequency.

osc o/p



deciding power as (f) increases.

• $S_{\phi n}(f)$ appears at 1 GHz freq o/p of osc.

• Each cycle = 1ns $\therefore$ (High-freq phase noise component can affect the phase (or zero crossing) $\rightarrow$ 1ns later)

✓ High freq components affect the Jitter by adjacent cycles (i.e., $T_{\text{cycle-to-cycle}}$)

high-pass filtering effect.

observing Jitter over one cycle
Low-freq noise do not change over that ^{small} time
Only high-freq noise comp $\rightarrow$ ~~filter~~ passed through.

white noise induced phase noise $S_{\phi n}(f)$

$$J_{CC, rms}^2 = \frac{4\pi}{\omega_0^3} \omega^2 S_{\phi n}(\omega)$$

$S_{\phi n}(\omega)$ is of form α/f^2

$$J_{CC, rms} = \sqrt{\frac{2\alpha}{f_0^3}}$$

$$T_0 = 1/f_0$$

$$\frac{J_{CC, rms}}{T_0} = \sqrt{\frac{2\alpha}{f_0}} \quad (4)$$

Example 2.15. 1GHz osc; phase noise = -90dBc/Hz; offset: 100kHz

a) J_{CC} ? if only $S_{\phi n}(f)$ is there (b) Jitter? after 50 cycles

$\rightarrow$ (a) $10 \log_{10} S_{\phi n}(f=100 \text{ KHz}) = -90 \text{ dBc/Hz}$ $\rightarrow J_{CC, rms} = \sqrt{\frac{20}{(1 \text{ GHz})^3}}$
 $\frac{\alpha}{(100 \text{ KHz})^2} = 10^{-9} \Rightarrow \alpha = 10 \text{ Hz}$ $J_{CC, rms} = 0.14$

(b) $J_N^2 = J_{1,2}^2 + J_{2,3}^2 + \dots + J_{49,50}^2 \rightarrow J_N = J_{50} = \sqrt{N} J_{CC, rms}$
 $J_{50} \approx 1 \text{ ps}$

Period Jitter $\rightarrow J_c \rightarrow$ Bounded random variations from the avg value T_{avg}
 $\rightarrow$ cycle jitter

$$J_c = \lim_{n \rightarrow \infty} \sqrt{\frac{1}{n} \sum_{k=1}^n (T_k - T_{avg})^2}$$

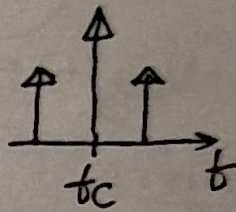
white noise induced jitter

$$J_c = \frac{J_{cc}}{\sqrt{2}}$$

Summary.

① Deterministic

due to periodic modulation

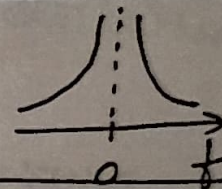


Bound

$$J_{pp} = \frac{q K_{vco} V_m}{\omega_m}$$

② Absolute Jitter

due to white noise

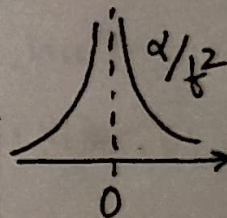


$$J_{rms} = \infty$$

unbound

③ Cycle-to-Cycle

due to white noise



bound

$$J_{rms} = \sqrt{\frac{2\alpha}{f_0^3}}$$

④ Period Jitter

due to white noise

same compared with T_{avg} instead.

$$J_c = \frac{J_{cc}}{\sqrt{2}}$$

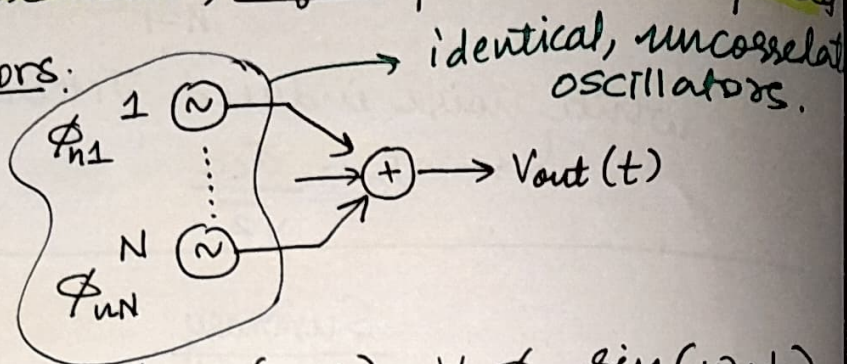
bound

2.3 Trade off b/w Phase Noise & Power.

- Direct trade off

- achieve lower phase noise, higher power in dissipated

■ Sum N-oscillators:



o/p: of 1 osc: $V_{out_1}(t) = V_0 \cos(\omega_1 t) - V_0 \phi_{n1} \sin(\omega_1 t)$

∴ N-oscillators.

$$V_{out}(t) = V_0 \cos(\omega_1 t) - V_0 \phi_{n1} \sin(\omega_1 t) + \dots + V_0 \cos(\omega_1 t) - V_0 \phi_{nN} \sin(\omega_1 t)$$

$\omega_1 \rightarrow$ average osc freq

$\phi_{n1} \dots \phi_{nN} \rightarrow$ phase noise of OSCS.

$$V_{out}(t) = N \cdot V_0 \cdot \cos(\omega_1 t) - V_0 (\phi_{n1} + \phi_{n2} + \dots + \phi_{nN}) \sin(\omega_1 t)$$

• normalised phase noises

$$\Rightarrow \frac{\phi_{n1} + \phi_{n2} + \dots + \phi_{nN}}{N}$$

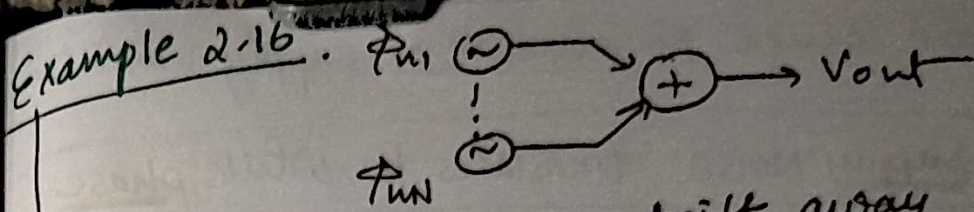
$$\Rightarrow S_{\phi}(f) = \frac{S_{\phi_{n1}}}{N} \leftarrow \text{phase noise spectra are equal.}$$

$$S_{\phi}(f) = \frac{S_{\phi_{n1}} + S_{\phi_{n2}} + \dots + S_{\phi_{nN}}}{N^2}$$

NOTE

✓ adding more identical osc reduces phase noise (N-increases)
But power also increases

doesn't reduce deterministic noise!!!



- Oscillators' phase drift away (altho they all start at the same value) over time.
- Won't lead to destructive addition of o/p voltage
- Will remain "Coherent" for thousands of cycles

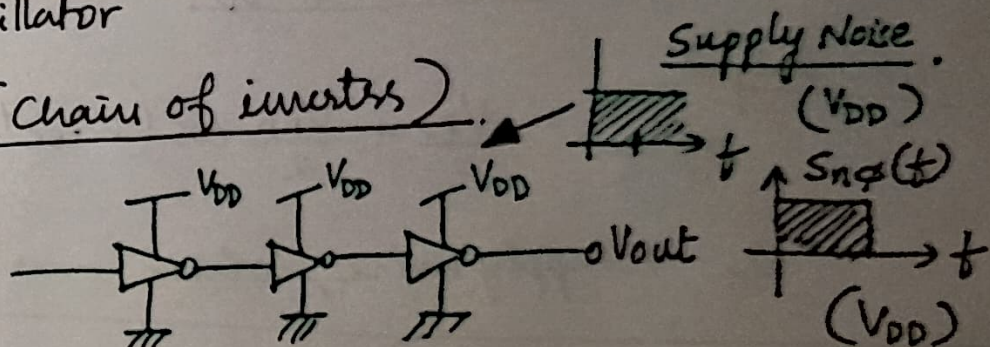
Q.4 BASIC NOISE MECHANISMS.

- phase vs freq noise
- Ring Osc.
- LC Osc.

Q.4.1 > Phase Noise vs freq noise

- Delayline
- Ring Oscillator

i) Delay line [Chain of inverters]



- delay of chain of inverters depend on supply voltage.
- ΔV in V_{DD} changes Δt in delay
- also a change in $\Delta\phi$ in input-output phase shift

Supply sensitivity / gain : $K_{VCO} = \frac{\Delta\phi}{\Delta V}$

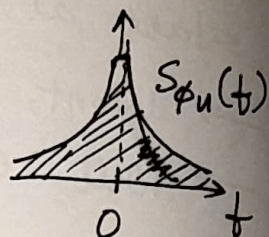
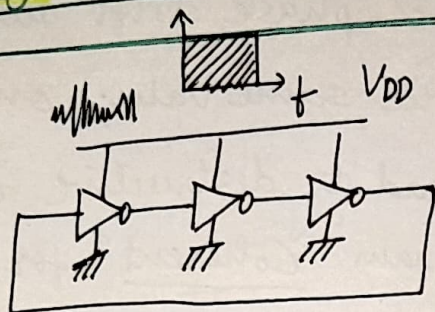
✓ Supply Noise, V_n

$$\begin{aligned} \rightarrow V_{out} &= \frac{V_{DD}}{2} + \frac{V_{DD}}{2} \cos(\omega_{int} t + \phi_0 + \Delta\phi) \\ &= \frac{V_{DD}}{2} + \frac{V_{DD}}{2} \cos(\omega_{int} t + \phi_0 + K_{VCO} \Delta V_n) \end{aligned}$$

$\phi_0 \rightarrow$ phase-shift in absence of supply variation

NOTE: white Supply Noise translates to white phase noise

(b) Ring Oscillator



$\rightarrow V_{DD}$ in ring osc. $\rightarrow$ affects both frequency & phase

$\rightarrow$ higher V_{DD} leads to higher osc frequency.

$\therefore$ white Supply Noise translates to white frequency noise

gain from V_{DD} : $K_{DD} = \frac{\partial \omega_{out}}{\partial V_{DD}}$ ^{freq component}

OSC output

$$V_{out} = \frac{V_{DD}}{2} + \frac{V_{DD}}{2} \cos(\omega_{osc} t + \int K_{DD} v_n dt)$$

$v_n \rightarrow$ Supply noise, white. $\rightarrow \frac{1}{f^2}$

if flicker noise: supply: $(1/f)$ phase noise in delay line
 $(1/f^3)$ in oscillators.

a & b
 * differ in spectral shape

Example 2.17. Infinitely long delay line Eg ring oscillator phase noise.

→ Noise sources in a delay line $\xrightarrow{\text{modulate}}$ delay $\downarrow$ phase.

✓ noise in each stage adds linearly as signal passes thro them ϕ_n o/p phase

✓ phase noise spectrum:

$$S_{\phi_n}(f) = \left(\frac{\alpha}{f} \right) + (\beta)$$

white noise (flat)

flicker noise (by $1/f$)

• Adding more inverters $\equiv$ increases Amplitude, power NOT specshape.

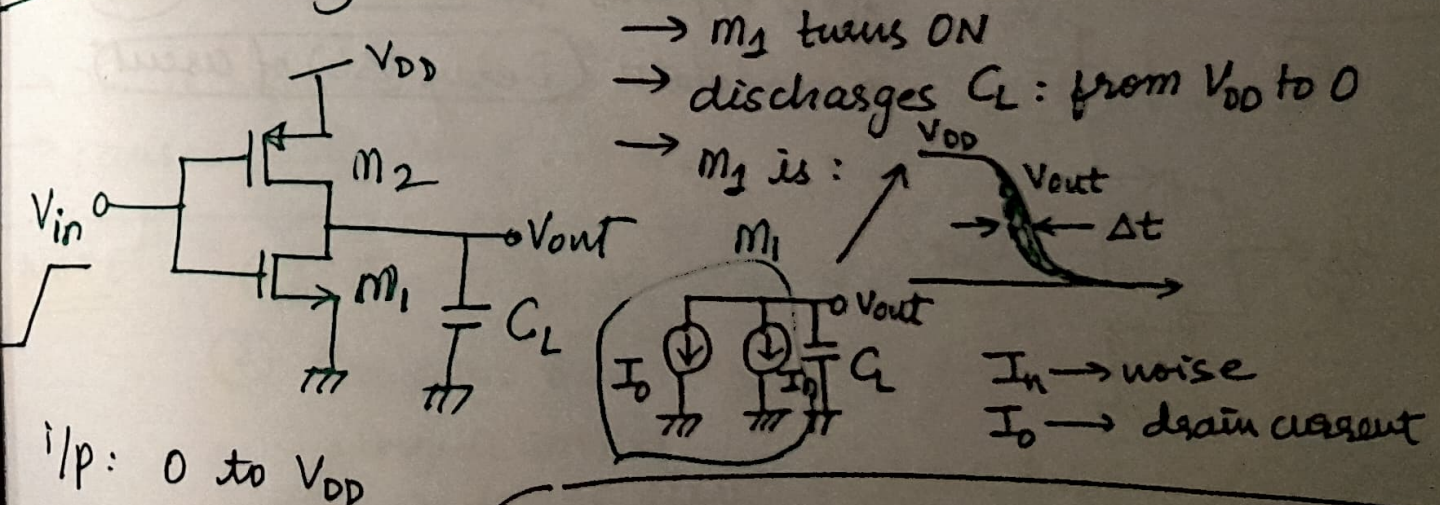
Oscillators.

phase noise spectrum $\Rightarrow S_{\phi_n}(f) = \left(\frac{\alpha}{f^2} \right) + \frac{\beta}{f}$

dominates.

feedback changes everything

2.4.2 Ring Oscillators.



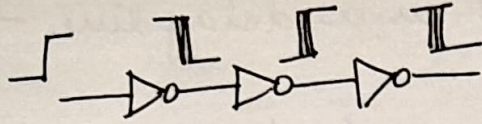
• flow of I_n through C_L leads to jitter Δt

• I_n has a white spectrum

• Δt also has white noise.

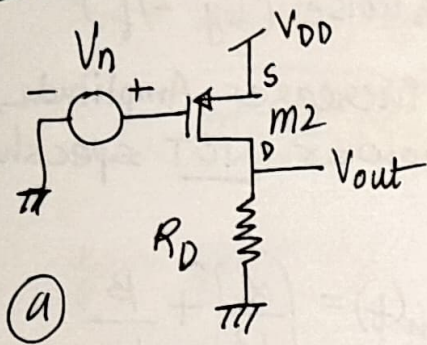
- I_n : white noise translates to white phase noise Δt

phase noise for delay line $\rightarrow (N)$ cascaded

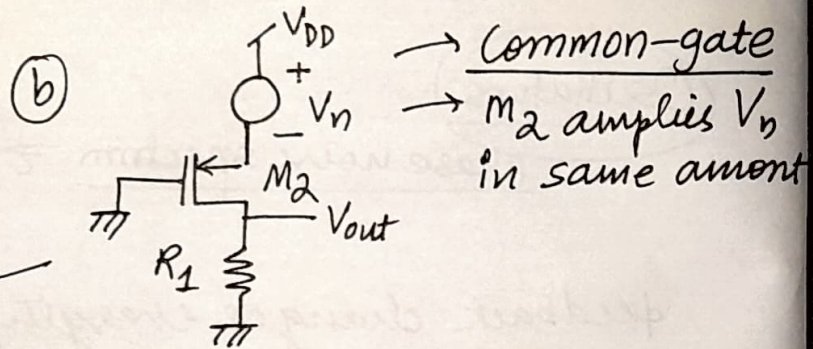


- phase noise rises by (N) factor
- rms jitter by $(\sqrt{N})$ factor.

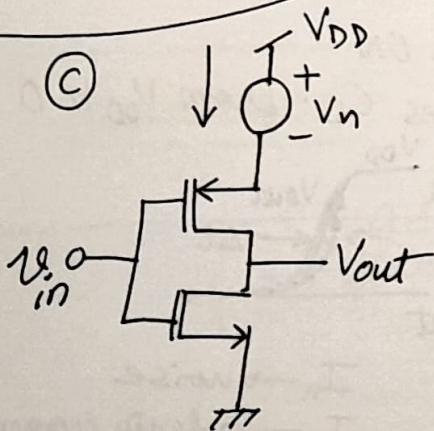
Example 2-18: flicker noise in PMOS.



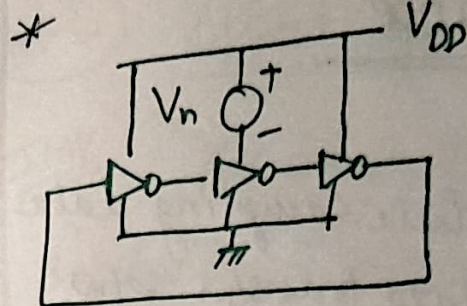
$\rightarrow M_2$ is a common-source device amplifies V_n .



Equivalent.



- (V_n) appears in series with (V_{DD})
- modulates Delay (Δt) of circuit



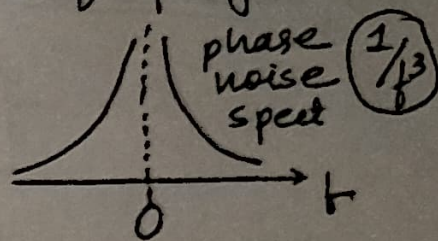
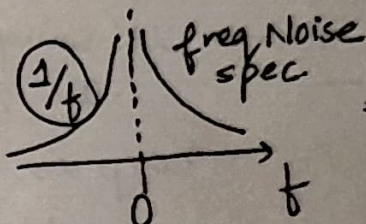
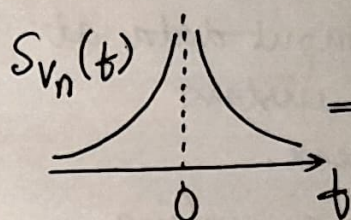
phase Noise???

→ V_n modulates Delay of 2nd invt

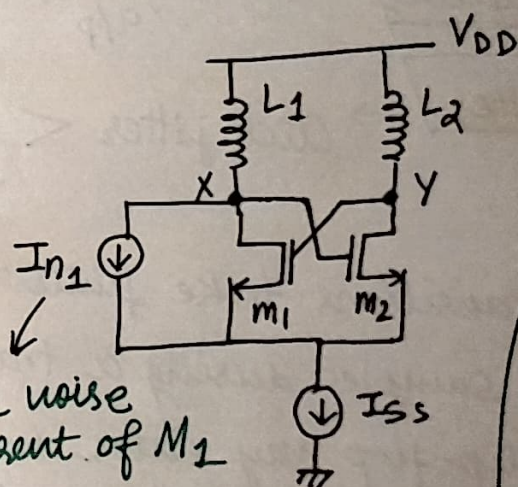
→ $f_0 \propto \frac{1}{\text{total delay}}$

∴ f_0 : OSC freq is modified

⇒ V_n → affects frequency noise



2.4.3 LC Oscillators



thermal noise current of M_1

→ I_{n1} is injected into (X).
→ causes Δt delay: Jitter.

NOTE

- ① $I_{n1} = 0$ when M_1 is OFF
- ② I_{n1} exists when M_1 is ON
- ③ I_{n1} doesn't flow from tank when M_2 is off
- (~~I_{n1} doesn't flow from tank~~)

→ I_{n1} is time-varying

NOTE: ① I_{n1} is zero; when M_1 is OFF.

② ~~the~~ other cycle: M_2 is OFF & M_1 is ON

→ loop is broken.

→ I_{n1} doesn't affect LC tank / doesn't flow from it.

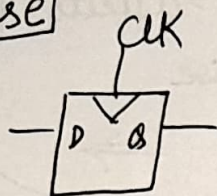
→ I_{SS} has an impedance r_o → attenuates I_{n1}

2.5 Effect of Jitter on Performance

① Jitter in Timing Systems

- Jitter may introduce errors while sampling data
- most crucial in high-speed data transfer b/w µp & memory.

② Ideal Case



- CLK samples input data at the ideal instant
- o/p is correct.

③ With Jitter

- at incorrect time → sampling
- causes logic errors.
- metastability : (ambiguous/incorrect)
o/p

Maximum Allowed Jitter

$$\text{clock jitter} < \frac{\text{data period}}{2}$$

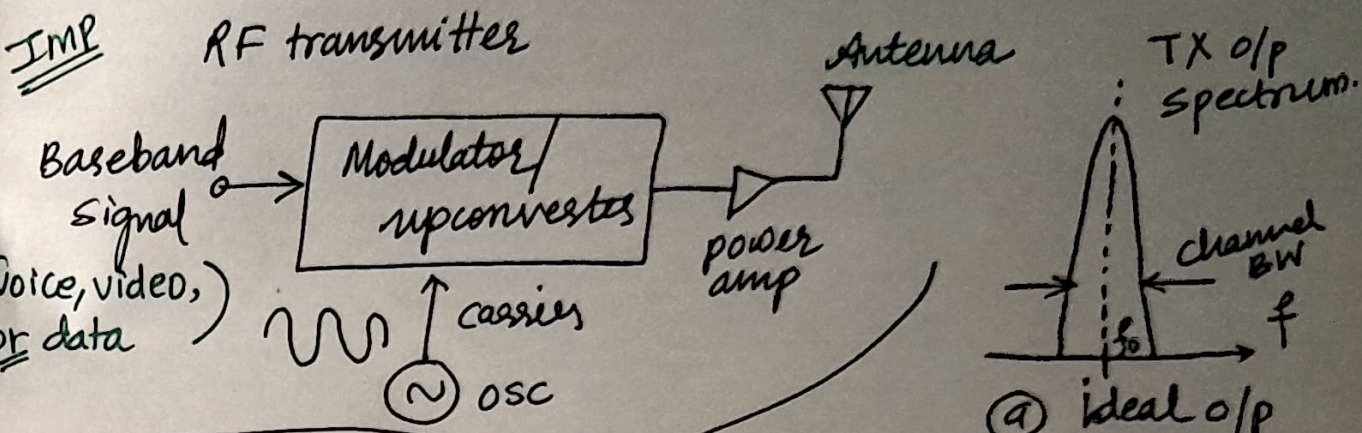
④ Real Case

- ① data transitions take finite time
- clock samples during a transition
- the flip-flop may enter metastability or interpret data wrongly.

Additional Issues

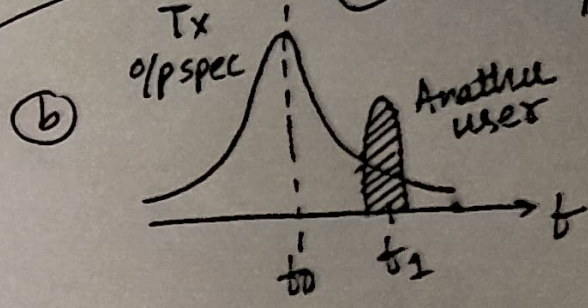
- Data can also have Jitter
- SKEW (b/w data & clock)
reduces allowable jitter margin.

2.6 Effect of Phase Noise ON Performance.



Ideal case

- no phase noise
- Tx output ⇒ $A(t) \cos(\omega_0 t + \phi(t))$
- ↓
- Carry baseband info.
- confined to specific channel BW (f_0) .



Weak Signals

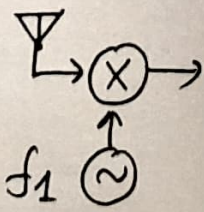
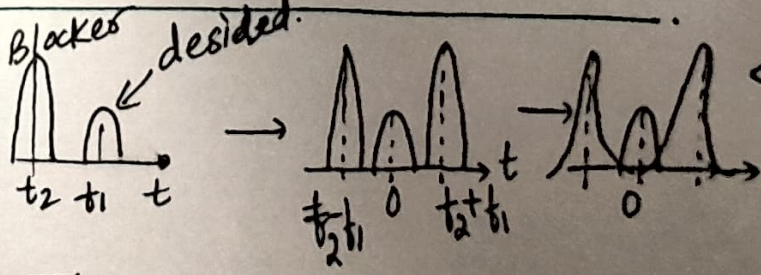
* Another user's signal at (f_1) is corrupted by high power signal at (f_0) .

phase noise corrupts weak signal.

Phase Noise.

$T_x: A(t) \cos(\omega_0 t + \phi(t) + \phi_n(t))$

Phase noise in Receivers.



← phase noise causes reciprocal mixing

- Rx local oscillator converts desired sig at f_1 to Baseband (voice, video)
- also downconverts Blockers.
- LO phase noise spreads into desired sig ✓ corrupts it
- BOTH TX & RX can CORRUPT!!